

JEE MAIN REVISION MASTER: PERIODIC TABLE AND PERIODICITY 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Ionization Enthalpy & Electron Gain Enthalpy	9 PYQs	Periodic Trends
Moderate	Atomic & Ionic Radii Trends	7 PYQs	Radii

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ionization Enthalpy & Electron Gain Enthalpy	$Z_{\text{eff}} = Z - \sigma, \quad \Delta_{\text{eg}}H(\text{Cl}) < \Delta_{\text{eg}}H(\text{F}) \quad \text{Cl is most negative in periodic table}$
Atomic & Ionic Radii Trends	$\text{For isoelectronic: } r \propto \frac{1}{Z} \implies \text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ionization Enthalpy & Electron Gain Enthalpy (High Yield — 9 PYQs)

Core Principle: Periodic variations of ionization energy, exceptions due to half-filled/fully-filled subshells ($\text{N} > \text{O}$, $\text{Be} > \text{B}$), and electron gain enthalpy.

■ **Caution / JEE Pitfall:** Ionization enthalpy of Nitrogen is HIGHER than Oxygen due to stable half-filled $2p^3$ subshell ($\text{N}: 2s^2 2p^3 > \text{O}: 2s^2 2p^4$).

■ **Speed Trick:** Chlorine has MORE negative electron gain enthalpy than Fluorine because of compact size and high inter-electronic repulsion in $2p$ of Fluorine.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Given below and the atomic masses of the elements: $\begin{matrix} \text{Element:} \\ \text{Li \& Na} \\ \text{\& Cl \& K \& Ca \& Br \& Sr \& I \& Ba} \end{matrix}$..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "A pair of amphoteric oxides is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Amphoteric oxide among the following..." [Ans: N/A]

3.2 Atomic & Ionic Radii Trends (Moderate Yield — 7 PYQs)

Core Principle: Covalent, van der Waals, and ionic radii trends across periods and groups, and size shrinkage in isoelectronic ions.

■ **Caution / JEE Pitfall:** Noble gas radii are measured as van der Waals radii, making them appear abnormally larger than adjacent halogens.

■ **Speed Trick:** Due to lanthanoid contraction (poor shielding of 4f electrons), 4d and 5d transition elements have nearly identical radii (e.g. Zr $\approx$ Hf).

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th group" **[Ans: N/A]**
- **[2023 Shift] Q#1 (MCQ):** "In which one of the following pairs, both the elements does not have $(n-1)d^1 10ns^2$ configuration in its elementary state?" **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "The correct order of first ionisation enthalpy of given elements is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th group (b) Ga – 31 (ii) 6th period, 6th group (c) W – 74 (iii) 5th period, 10th group (d) Pd – 46 (iv) 7th period, 2nd group Choose the correct answer from the options given below.

- (A) a-iv, b-i, c-ii, d-iii
- (B) a-i, b-ii, c-iii, d-iv
- (C) a-iv, b-ii, c-iii, d-i
- (D) a-iii, b-iv, c-i, d-ii

Correct Answer: None

Solution Analysis: For finding the position of an element in the periodic table, we use its electronic configuration. (a) Ra, atomic number 88 Ra belongs to alkaline earth metals. Its outer configuration is: Ra: [Rn]7s² Highest shell number is 7, so it is in the 7th period. It has 2 valence electrons in the s-subshell, so it is in the 2nd group. So, Ra matches with (iv). (b) Ga, atomic number 31 Electronic configuration: Ga: [Ar]3d¹⁰4s²4p¹ Highest shell number is 4, so it is in the 4th period. It has configuration ns²np¹, so it belongs to group 13.

Benchmark Problem 2: KCET 2023 (Q#4)

A pair of amphoteric oxides is

- (A) Al₂O₃, Li₂O
- (B) BeO, BO₃
- (C) BeO, MgO
- (D) BeO, ZnO

Correct Answer: None

Solution Analysis: BeO, Al₂O₃ and ZnO are amphoteric in nature MgO and Li₂O are basic in nature whereas BO₃ is acidic. Thus, the pair of amphoteric oxides are BeO and ZnO.